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# RESONANT MOTION OF A SPHERICAL PENDULUM

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The weakly nonlinear, resonant response of a damped, spherical pendulum (length $l$, damping ratio $\delta$, natural frequency $\omega_0$) to the planar displacement $\epsilon l \cos \omega t$ ($\epsilon \ll 1$) of its point of suspension is examined in a four-dimensional phase space in which the coordinates are slowly varying amplitudes of a sinusoidal motion. The loci of equilibrium points and the corresponding bifurcation points in this space are determined. The control parameters are $\alpha = 2\delta/\epsilon^{2/3}$ and $\nu = 2(\omega^2 - \omega_0^2)/\epsilon^{2/3}\omega^2$. If $\alpha < 0.441$ there is a finite interval of $\nu$ within which no stable equilibrium points exist. As $\nu$ decreases through the upper bound (a Hopf-bifurcation point) of this interval the motion in the phase space becomes periodic and then, following a period-doubling cascade, chaotic. There may be alternating sub-intervals of chaotic and periodic motion. The chaotic trajectories in the phase space appear to lie on fractal attractors.

## 1. Introduction

Some twenty years ago [1] I examined the weakly nonlinear response of a lightly damped, spherical pendulum to a simple harmonic, planar displacement of the point of suspension. My primary concern at that time was the possible excitation of non-planar motion in the neighborhood of resonance (forcing frequency ≈ natural frequency). I concluded that for sufficiently small damping: (a) planar harmonic† motion is unstable in a certain neighborhood of the natural frequency; (b) non-planar harmonic motion is stable in a spectral neighborhood that overlaps neighborhoods of both stable and unstable planar motions; and (c) no stable, harmonic motion is possible in a finite neighborhood of the natural frequency. The aharmonic motions are modulated sinusoids, the slowly varying amplitudes of which are governed by four, first-order, autonomous, nonlinear, ordinary differential equations (n.o.d.e.). The solutions of these n.o.d.e. were determined on an analog computer and, although rather complex, appeared to be periodic (but no power spectra were computed).

The spherical pendulum is perhaps the simplest example of a two-degree-of-freedom oscillator with identical natural frequencies. The two degrees of freedom are uncoupled in the linear approximation but are resonantly coupled by nonlinearity. Similar formulations may be obtained for the resonantly forced motion of any pair of nonlinearly coupled, weakly damped oscillators of approximately equal natural frequencies provided that the nonlinearity is weak and comprises a cubic component (a quadratic component would not contribute to resonant coupling and therefore would be negligible in the present context). A problem of some practical importance in this category is posed by the free-surface motion of a liquid in a moving circular tank, which is analogous to that of the spherical pendulum if only the two dominant modes (of equal frequency and orthogonal nodal lines) are retained in the solution of the equations of motion [2].

Not long after [1] was completed, Lorenz [3] discovered that the solution of a set of three, first-order, autonomous n.o.d.e. may be chaotic

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†I use "harmonic" to mean *simple harmonic* or, equivalently, *monochromatic*.

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(even though the forcing is regular), a phenomenon that has since been widely studied [4]. It therefore appears worthwhile to revisit the pendulum problem, which is more complex than Lorenz's problem in being governed by four, rather than three, n.o.d.e. but is simpler in comprising only two, rather than three, parameters.

These two parameters, which measure damping and proximity to resonance, are subsequently taken to be

$$\alpha = 2\delta\varepsilon^{-2/3}(\omega_0/\omega) \tag{1.1}$$

and

$$\nu = \varepsilon^{-2/3}(\omega^2 - \omega_0^2)/\omega^2, \tag{1.2}$$

where $\delta$ is the damping ratio (prescribed damping/critical damping for small oscillations), $\varepsilon$ is the ratio of the maximum displacement of the point of suspension to the length of the pendulum, $\omega$ is the forcing frequency, and $\omega_0$ is the natural frequency. I assume that $0 < \varepsilon \ll 1$ (weak nonlinearity), $\alpha = \mathcal{O}(1)$, and $\nu = \mathcal{O}(1)$; nonlinearity is insignificant, and the pendulum exhibits only planar harmonic motion, if $\varepsilon$ is small and either $\alpha$ or $|\nu|$ is (or both are) large. The factor $\omega_0/\omega$ in (1.1), which could be replaced by 1 within the parametric domain $\nu = \mathcal{O}(1)$, has been included to render the subsequent formulation uniformly valid for all $\omega$ in the limit $\varepsilon \downarrow 0$.

Let $E$ be a measure of the total energy of the motion and $E = E(\nu)$ be the locus, not necessarily single valued, of equilibrium (fixed) points of the n.o.d.e. for a prescribed value of $\alpha$. A bifurcation point then separates stable from unstable points on such a locus (or *resonance curve*†). The number of bifurcation points, which are determined and catalogued in sections 3 and 4 below, is 2 for $0.625 > \alpha > 0.526$, 3 for $0.526 > \alpha > 0.441$, 5 for $0.441 > \alpha > 0.433$, and 7 for $0.433 > \alpha > 0$ (three of the bifurcation points recede to $\nu = -\infty$ and one to $\nu = +\infty$ as $\alpha \downarrow 0$). Forced, planar oscillations of the pendulum are stable for all $\nu$ (there are no bifurcation points) if $\alpha > 0.625$. Either planar or non-planar harmonic motion is stable in contiguous but non-overlapping ranges of $\nu$ if $0.625 > \alpha > 0.526$; this remains true in $0.526 > \alpha > 0.441$, except that there then is an overlapping range of planar and non-planar stability (there is an additional overlapping range of planar and non-planar stability if $\alpha < 0.433$). Two Hopf-bifurcation points (all other bifurcation points are of Poincaré's type) appear for $\alpha < 0.441$, and no stable harmonic motion is possible in the spectral range bounded by these two points if $0.441 > \alpha > 0.374$. This range of no stable harmonic motion remains for $\alpha < 0.374$, but its lower bound (minimum $\nu$) then is a Poincaré-, rather than a Hopf-, bifurcation point.

The trajectories in the parametric domain of no stable harmonic motion prove to be either periodic or chaotic. If $0 < \nu_{2*} - \nu \ll 1$, where $\nu_{2*} = \nu_{2*}(\alpha)$ is the upper Hopf-bifurcation point, the motion in the phase space is approximately harmonic (the projection of the trajectory, which then is a limit cycle, on a plane in the phase space is approximately elliptical). This motion remains periodic (the projection of the trajectory on a plane in the phase space is a smooth, closed curve) for sufficiently small $\nu_{2*} - \nu$ but exhibits a sequence of period doublings with decreasing $\nu$ and then becomes chaotic for some critical value of $\nu$ if $\alpha$ is sufficiently small. A further decrease in $\nu$ leads again to periodic motion, and, depending on the value of $\alpha$, there may be alternating sub-intervals of periodic and chaotic motion in the interval $\nu_3 < \nu < \nu_{2*}$, where $\nu_3 = \nu_3(\alpha)$ is the lower bound of the spectral range of no stable harmonic motion. The chaotic trajectories appear to lie on fractal (strange) attractors† of the types obtained by Lorenz [3, 5] and Rössler [6].

The reader who wishes to pass over the details of the bifurcation points may go directly from the

†The resonance curves in [1] are plotted in an $a$–$\nu$ plane, where $a$ is a measure of the pendulum-bob displacement in the plane of excitation ($a = \sqrt{E}$ for planar, but not for non-planar, motion). The fact that the non-planar resonance curve might have two, non-contiguous stable segments was overlooked in [1], as also was the bifurcation point $\nu_{1*}$ (see figs. 1 and 2).

†Following Mandelbrot [7], I have used the adjective *fractal* in preference to *strange*.

formulation of the phase-space equations in section 2 to the discussion of their numerical solutions in sections 5 and 6.

## 2. Phase-space equations

We consider a simple pendulum of length $l$ and mass $m$ (damping is introduced subsequently). The Cartesian coordinates of the point of suspension and the bob are $(x_0, 0, 0)$, and $(x, y, z)$, respectively, where

$$x_0 = \varepsilon l \cos \omega t, \tag{2.1}$$

and satisfy the constraint

$$(x - x_0)^2 + y^2 + z^2 = l^2. \tag{2.2}$$

The Lagrangian is

$$L = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - m\omega_0^2 l(l - z), \tag{2.3}$$

where $\dot{x} \equiv \mathrm{d}x/\mathrm{d}t, \ldots,$ and

$$\omega_0 = (g/l)^{1/2} \tag{2.4}$$

is the natural frequency (for small displacements) of the pendulum.

We assume that $\omega^2$ approximates $\omega_0^2$ according to

$$\omega^2 = \omega_0^2 + \varepsilon^{2/3}\omega^2 \nu, \tag{2.5}$$

where $\nu = \mathcal{O}(1)$ as $\varepsilon \downarrow 0$, and pose the solution in the form

$$x = \varepsilon^{1/3} l[\, p_1(\tau)\cos\theta + q_1(\tau)\sin\theta\,] \tag{2.6a}$$

and

$$y = \varepsilon^{1/3} l[\, p_2(\tau)\cos\theta + q_2(\tau)\sin\theta\,], \tag{2.6b}$$

where

$$\theta = \omega t, \quad \tau = (1/2)\varepsilon^{2/3}\omega t, \tag{2.7a,b}$$

and $p_1, \ldots$ are slowly varying coordinates in a four-dimensional phase space† ($p_1, q_1, p_2, q_2 \equiv f_{1,2,3,4}$ in [1]). The trajectory described by (2.6), which is the projection of the pendulum-bob trajectory on a horizontal $(x, y)$ plane, is, in general, a slowly rotating ellipse with slowly varying axes (it is a fixed ellipse at an equilibrium point, at which $p_1, \ldots$ are fixed); see appendix A for details.

Substituting (2.6) into (2.3), averaging the resulting Lagrangian over $\theta$, invoking (2.5), retaining only the dominant terms in the limit $\varepsilon \downarrow 0$, and requiring the average Lagrangian to be stationary with respect to variations of $p_1, \ldots$ (Hamilton's principle), we find that the resulting evolution equations may be placed in the Hamiltonian form

$$\frac{\mathrm{d}p_i}{\mathrm{d}\tau} = -\frac{\partial H}{\partial q_i}, \quad \frac{\mathrm{d}q_i}{\mathrm{d}\tau} = \frac{\partial H}{\partial p_i} \quad (i = 1, 2), \tag{2.8a,b}$$

where

$$H(p_1, q_1, p_2, q_2) = p_1 + \frac{1}{2}\nu E + \frac{1}{32}E^2 - \frac{3}{8}M^2, \tag{2.9}$$

$$E = p_1^2 + q_1^2 + p_2^2 + q_2^2 \equiv R^2, \quad M = p_1 q_2 - p_2 q_1. \tag{2.10a,b}$$

We remark that $E$ is a measure of the energy of the pendulum, $R$ is both the hyper-radius in the $p_1$–$q_1$–$p_2$–$q_2$ space and a measure of the r.m.s. displacement

$$\left[\frac{1}{2\pi}\int_0^{2\pi}(x^2 + y^2)\,\mathrm{d}\theta\right]^{1/2} = 2^{-1/2}\varepsilon^{1/3} l R, \tag{2.11}$$

and $M$ is a measure of the angular momentum. (The angular momentum is referred to the vertical axis through $x = y = 0$ and is generated by the

†The four, first-order n.o.d.e. for $x$, $\dot{x}$, $y$ and $\dot{y}$ vs. $t$, as determined by (2.2) and (2.3), are non-autonomous but may be rendered autonomous by introducing the fifth variable $\theta = \omega t$ and the fifth equation $\dot{\theta} = \omega$. The $p_1$–$p_2$–$q_1$–$q_2$ phase-space trajectories are Poincaré maps (surfaces of section [8]) of the $x$–$\dot{x}$–$y$–$\dot{y}$–$\theta$ trajectories at $\theta = 0 \pmod{2\pi}$. The $p_1$–$p_2$ and $q_1$–$q_2$ projections of the $p_1$–$q_1$–$p_2$–$q_2$ trajectories are maps of the $x$–$y$ pendulum trajectories at the periodically spaced instants $\omega t = 0 \pmod{2\pi}$ and $1/2\pi \pmod{2\pi}$, respectively.

$y$-component of the reaction force at the point of support.)

Now suppose that the pendulum is lightly damped, such that the ratio of the damping to critical damping for plane motion is given by

$$\delta = (1/2)\varepsilon^{2/3}(\omega/\omega_0)\alpha, \tag{2.12}$$

where $\alpha = \mathcal{O}(1)$ as $\varepsilon \downarrow 0$. The terms $-\alpha p_i$ and $-\alpha q_i$ then must be added to the right-hand sides of (2.8a, b), respectively. The end results are

$$\dot{p}_1 = -\alpha p_1 - \left(\nu + \frac{1}{8}E\right)q_1 - \frac{3}{4}Mp_2, \tag{2.13a}$$

$$\dot{q}_1 = -\alpha q_1 + \left(\nu + \frac{1}{8}E\right)p_1 - \frac{3}{4}Mq_2 + 1, \tag{2.13b}$$

$$\dot{p}_2 = -\alpha p_2 - \left(\nu + \frac{1}{8}E\right)q_2 + \frac{3}{4}Mp_1, \tag{2.13c}$$

$$\dot{q}_2 = -\alpha q_2 + \left(\nu + \frac{1}{8}E\right)p_2 + \frac{3}{4}Mq_1, \tag{2.13d}$$

where the dots now imply differentiation with respect to $\tau$. Note that (2.13) are invariant under the reflection $(p_2, q_2) \rightarrow -(p_2, q_2)$ by virtue of the invariance of (2.2) and (2.3) under the reflection $y \rightarrow -y$.

The divergence of the set (2.13) in the four-dimensional phase space is

$$\Delta \equiv \frac{\partial \dot{p}_1}{\partial p_1} + \frac{\partial \dot{q}_1}{\partial q_1} + \frac{\partial \dot{p}_2}{\partial p_2} + \frac{\partial \dot{q}_2}{\partial q_2} = -4\alpha, \tag{2.14}$$

from which it follows that an element of volume in the phase space contracts like $\exp(-4\alpha\tau)$, and every trajectory ultimately must be confined to a limiting subspace of dimension less than four (cf. [9]). It also may be shown that every trajectory ultimately must lie in $R \leq 1/\alpha$ (appendix B).

## 3. Equilibrium points

We obtain the equilibrium (also *fixed* or *singular*) points of (2.13) by setting $\dot{p}_i = \dot{q}_i = 0$ and solving for $p_i$ and $q_i$ $(i = 1, 2)$. It is convenient to express $p_i$ and $q_i$ in terms of $E$ and $M$ and then solve for $E$ and $M$. Multiplying (2.13a, b, c, d) through by $(p_1, q_1, p_2, q_2)$ and $(q_2, -p_2, -q_1, p_1)$, respectively, and summing the results, we obtain

$$q_1 = \alpha E, \quad p_2 = -2\alpha M. \tag{3.1a, b}$$

Substituting (3.1a, b) into (2.13a, d) and solving for $p_1$ and $q_2$ on the assumption that $\alpha \neq 0$, we obtain

$$p_1 = \frac{3}{2}M^2 - \left(\nu + \frac{1}{8}E\right)E, \quad q_2 = \left(\frac{1}{2}E - 2\nu\right)M. \tag{3.1c, d}$$

Substituting (3.1) into (2.13c), we obtain either

$$M = 0 \quad \text{or} \tag{3.2a}$$

$$M^2 = \frac{5}{36}\left[E^2 + \frac{32}{5}\nu E - \frac{64}{5}(\nu^2 + \alpha^2)\right]. \tag{3.2b}$$

Substituting (3.1) into (2.13b) and invoking either (3.2a) or (3.2b) to eliminate $M$, we obtain either

$$E^3 + 16\nu E^2 + 64(\nu^2 + \alpha^2)E - 64 = 0, \tag{3.3a}$$

or

$$E^3 + 16(\alpha^2 - 3\nu^2)E + 128\nu(\nu^2 + \alpha^2) - 24 = 0, \tag{3.3b}$$

which correspond to planar or non-planar motion, respectively. Only the positive real roots of (3.3a) and those positive real roots of (3.3b) for which $M^2 \geq 0$ are physically significant.

### 3.1. *Planar motion*

It is expedient to solve (3.3a) for $\nu = \nu(E)$, rather the $E = E(\nu)$, and then to plot the amplitude $\sqrt{E} \equiv R$ versus the frequency parameter $\nu$ with $\alpha$ as a family parameter. The end results are

$$\nu = -\tfrac{1}{8}E \mp E^{-1/2}(1 - \alpha^2 E)^{1/2}, \tag{3.4}$$

$$p_1 = \pm E^{1/2}(1 - \alpha^2 E)^{1/2}, \tag{3.5a}$$

$$q_1 = \alpha E \quad (p_2 = q_2 = 0), \tag{3.5b}$$

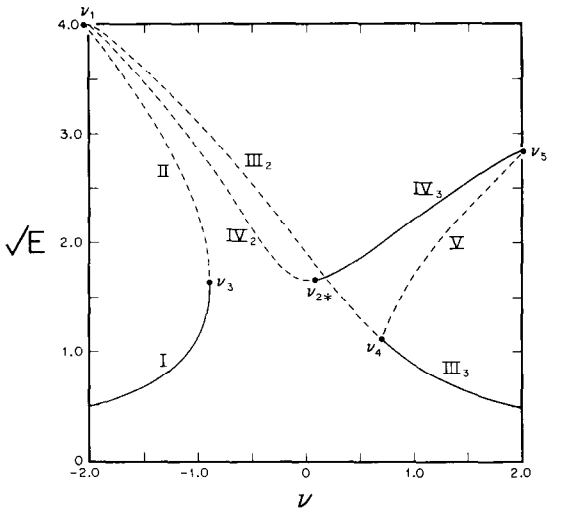



Fig. 1. The resonance curves (loci of equilibrium points) for $\alpha = \frac{1}{4}$. The resonance curves for $0 < \alpha < 0.433$ are qualitatively similar except that $\nu_{1*} > \nu_3$ for $\alpha > 0.374$. The branches depicted by solid/dashed lines are stable/unstable. The stable branch $III_1$, which extends from the point of vertical tangency on the left ($\nu = \nu_1 = -2.008$) to the intersection with the stable branch $IV_1$ on the right ($\nu = \nu_2 = -1.994$), and the stable branch $IV_1$, which extends from this intersection to $\nu = \nu_{1*} = -1.986$, are not resolved on the scale of the drawing (cf. fig. 2).

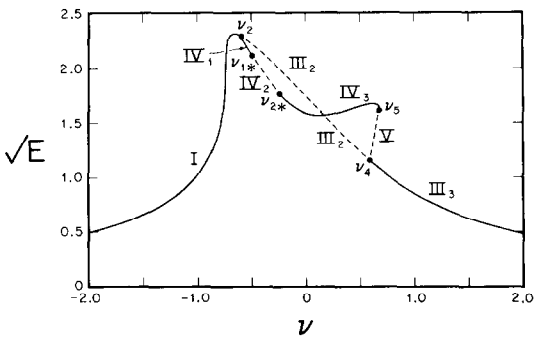



Fig. 2. The resonance curves for $\alpha = 0.433$. The resonance curves for $0.433 < \alpha < 0.441$ are qualitatively similar. Those for $0.441 < \alpha < 0.526$ differ only in that the entire upper branch of the non-planar resonance curve is stable (branch $IV_2$ disappears for $\alpha > 0.441$).

wherein the alternative signs are vertically ordered. See figs. 1-4.

Let $\nu_1$ and $\nu_3$ be the minimum and maximum values of $\nu(E)$ (the indices are chosen in anticipation of the subsequent development, in which the

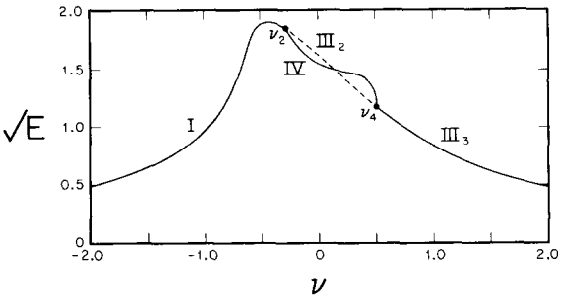



Fig. 3. The resonance curves for $\alpha = 0.526$. The resonance curves for $0.526 < \alpha < 0.625$ are qualitatively similar.

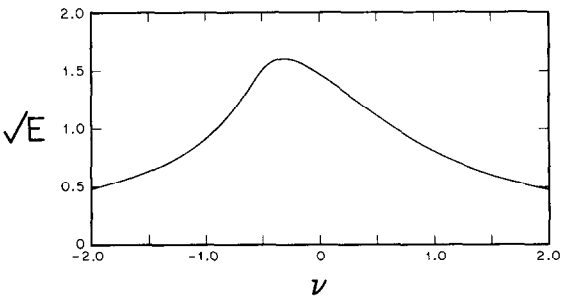



Fig. 4. The resonance curve for $\alpha = 0.625$. The resonance curves for $\alpha > 0.625$ are qualitatively similar.

bifurcation points $\nu = \nu_n$ are ordered according to $\nu_n < \nu_{n+1}$ for sufficiently small $\alpha$). The planar resonance curve $R = R(\nu)$ then has three branches if $\nu_1 < \nu < \nu_3$ but only one branch if either $\nu < \nu_1$ or $\nu > \nu_3$. We designate the latter branches and their continuations into $\nu > \nu_1$ and $\nu < \nu_3$ as I and III, respectively, and the intermediate branch as II† (fig. 1; see section 4 regarding the subdivision of III into stable and unstable segments). The resonant peak, which occurs on branch III, is given by

$$E_{\max} = \alpha^{-2} \quad \text{at} \quad \nu = -\tfrac{1}{8}\alpha^{-2}. \qquad (3.6a, b)$$

The upper/lower choice of signs in (3.4) and (3.5) applies to the left/right of this maximum on branch III; the upper choice applies on both I and II.

Requiring the derivative of (3.4) with respect to

†The present numbering of the branches differs from that in [1].

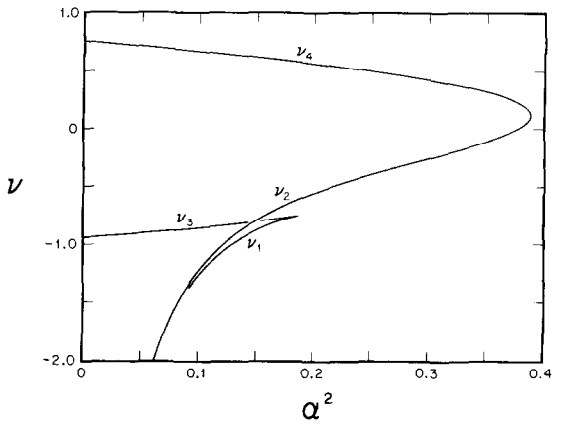



Fig. 5. The bifurcation points $\nu_1$ and $\nu_3$, as determined by (3.7) and (3.4) with the upper choice of sign, and $\nu_2$ and $\nu_4$, as determined by (3.12) and (3.4) with the lower choice of sign.

$E$ to vanish, we obtain

$$E^3(1-\alpha^2 E)=16, \tag{3.7}$$

which has two positive real roots for sufficiently small $\alpha$. The substitution of these two roots into (3.4) with the upper choice of signs yields $\nu_1$ and $\nu_3$, which are plotted in fig. 5. For $\alpha^2 \ll 1$,

$$\nu_1 = -\frac{1}{8}\alpha^{-2} - 2\alpha^4 + \mathcal{O}(\alpha^{10}), \tag{3.8a}$$

$$\nu_3 = -0.945 + 0.794\alpha^2 + \mathcal{O}(\alpha^4). \tag{3.8b}$$

The coincidence of the positive real roots of (3.7) implies

$$\nu_1 = \nu_3 = -3/4, \quad \text{for } \alpha = \sqrt{3/4} = 0.433; \tag{3.9}$$

(3.7) has no positive real roots, branch II disappears, and the planar resonance curve is single valued if $\alpha > 0.433$.

### 3.2. *Non-planar motion*

The non-planar resonance curve is determined by (3.3b) subject to the side condition $M^2(3.2b) > 0$. Note that each point on this curve corresponds to a pair of equilibrium points, which are related through $(M, p_2, q_2) \rightarrow -(M, p_2, q_2)$ [cf. remark following (2.13)]. Let $\nu_2$ and $\nu_4$ be those values of $\nu$ (we anticipate that there are at most two such values) at which $M^2$ changes sign on the curve determined by (3.3b) and at which (since (3.3a) and (3.3b) must be equivalent for $M = 0$) the non-planar resonance curve terminates on the planar resonance curve†, and let $\nu_5$ be the maximum value of $\nu = \nu(E)$ or, equivalently, that value of $\nu$ at which (3.3b) has a significant $(E, M^2 > 0)$ double root. It can be shown that $\nu_5 > \nu_4$ for sufficiently small $\alpha$ and that the minimum value of $\nu(E)$ corresponds to $M^2 < 0$ and therefore is irrelevant in the present context. The non-planar resonance curve then is confined to $\nu_2 < \nu < \nu_5$ and has either one branch if $\nu_2 < \nu < \nu_4$ or two branches if $\nu_4 < \nu < \nu_5$ (see figs. 1–3). We designate the first branch and its continuation as the upper of the latter two branches as IV (see section 4 regarding the subdivision of IV into stable and unstable segments) and the lower of the latter two as V. The maximum and the minimum of $E = E(\nu)$ both lie on IV and are given by

$$E_{\max} = \frac{1}{2}\alpha^{-2} + \frac{32}{9}\alpha^4 + \mathcal{O}(\alpha^{10}) \tag{3.10a}$$

$$\text{at } \nu = \frac{1}{8}\alpha^{-2} - \frac{16}{9}\alpha^4 + \mathcal{O}(\alpha^{10}) \tag{3.10b}$$

and

$$E_{\min} = 2.885 - 1.849\alpha^2 + \mathcal{O}(\alpha^4) \tag{3.11a}$$

$$\text{at } \nu = 0.462\alpha^2 + \mathcal{O}(\alpha^4). \tag{3.11b}$$

It can be shown that the commonality of the roots of (3.3b) and $M^2(3.2b) = 0$ implies (through the elimination of $\nu$)

$$E^3(1-\alpha^2 E) = 16/9, \tag{3.12}$$

which has two positive real roots for sufficiently small $\alpha^2$. The substitution of these roots into (3.4) with the lower choice of sign yields the bifurcation points $\nu_2$ and $\nu_4$, which are plotted in fig. 5. For

†The intersection of the non-planar branch IV with the planar branch III is not a bifurcation point since $M^2 > 0$ there.

$\alpha^2 \ll 1$,

$$\nu_2 = -\frac{1}{8}\alpha^{-2} + \frac{14}{9}\alpha^4 + \mathcal{O}(\alpha^{10}), \tag{3.13a}$$

$$\nu_4 = 0.757 - 0.795\alpha^2 + \mathcal{O}(\alpha^4). \tag{3.13b}$$

The bifurcation point $\nu_5$, at which $\nu(E)$ has a maximum, is determined by the larger positive root of

$$\left(\nu^2 - \frac{1}{3}\alpha^2\right)^{3/2} = \nu(\nu^2 + \alpha^2) - \frac{3}{16} \tag{3.14}$$

and is plotted in fig. 6. If $\alpha^2 \ll 1$

$$\nu_5 = \frac{1}{8}\alpha^{-2} + \frac{2}{9}\alpha^4 + \mathcal{O}(\alpha^{10}). \tag{3.15}$$

The coincidence of the maximum of $\nu = \nu(E)$ with $M = 0$ implies

$$\nu_5 = \nu_4 = 0.470 \text{ and } E = 1.433 \quad \text{for } \alpha = 0.526; \tag{3.16}$$

the bifurcation point $\nu_5$ and branch V disappear for $\alpha > 0.526$. The coincidence of the positive real roots of (3.12) implies

$$\nu_2 = \nu_4 = 0.120 \text{ and}$$
$$E = 1.924 \quad \text{for } \alpha = 3^{5/6}/4 = 0.625; \tag{3.17}$$

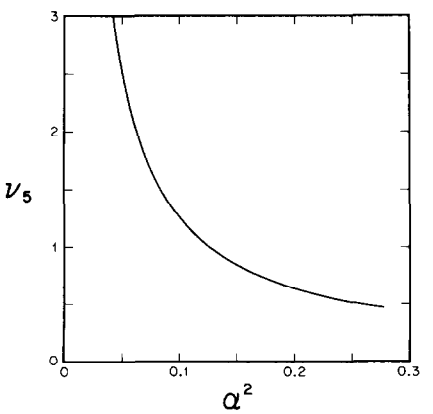



Fig. 6. The bifurcation point $\nu_5$, as determined by (3.14). The curve terminates at $\alpha^2 = 0.277$, where $\nu_5 = \nu_4 = 0.470$.

(3.12) has no positive real roots, and the non-planar resonance curve disappears, if $\alpha > 0.625$.

The resonance curves for $\alpha = 1/4$, 0.433, 0.526 and 0.625 are plotted in figs. 1–4. The neighborhood of the planar maximum and the intersection of $III_1$, $III_2$ and $IV_1$, which is not resolved on the scale of fig. 1, may be examined by expanding (3.3 a, b) about $E = \alpha^{-2}$ and $\nu = -1/8\alpha^{-2}$. The end results are

$$E = \alpha^{-2} + 32\alpha^4\left[\pm(1 + \tfrac{1}{2}\bar{\nu})^{1/2} - 1 - \tfrac{1}{4}\bar{\nu}\right] + \mathcal{O}(\alpha^{10}), \tag{3.18a}$$

and

$$E = \alpha^{-2} + \frac{32}{3}\alpha^4\left(1 - \frac{3}{4}\bar{\nu}\right) + \mathcal{O}(\alpha^{10}) \quad \left(\bar{\nu} > \frac{14}{9}\right) \tag{3.18b}$$

where

$$\bar{\nu} = \alpha^{-4}\left(\nu + \frac{1}{8}\alpha^{-2}\right), \tag{3.19}$$

the lower/upper choice of sign in (3.18a) corresponds to $II/III_{1,2}$, and (3.18b) corresponds to $IV_1$.

## 4. Stability of equilibrium points

Substituting

$$(p_i, q_i) = (p_i^0, q_i^0) + (P_i, Q_i)\,e^{\lambda\tau}, \tag{4.1}$$

where $(p_i^0, q_i^0)$ are the coordinates of a particular equilibrium point and $|P_i|$, $|Q_i| \ll 1$ ($\lambda$ in [1] is $\lambda + \alpha$ in the present notation), into (2.13) and requiring the determinant of the resulting linear equations in $P_i$ and $Q_i$ to vanish, we obtain

$$F(\lambda) = (\lambda + \alpha)^4 + D_2(\lambda + \alpha)^2 + D_0 = 0, \tag{4.2}$$

where

$$D_0 = \begin{vmatrix} \frac{1}{4}p_1q_1 + \frac{3}{4}p_2q_2 & \hat{\nu} + \frac{1}{4}q_1^2 - \frac{3}{4}p_2^2 & \frac{3}{4}p_1q_2 - \frac{5}{4}p_2q_1 & \frac{3}{4}p_1p_2 + \frac{1}{4}q_1q_2 \\ -\hat{\nu} - \frac{1}{4}p_1^2 + \frac{3}{4}q_2^2 & -\frac{1}{4}p_1q_1 - \frac{3}{4}p_2q_2 & -\frac{1}{4}p_1p_2 - \frac{3}{4}q_1q_2 & \frac{5}{4}p_1q_2 - \frac{3}{4}p_2q_1 \\ -\frac{5}{4}p_1q_2 + \frac{3}{4}p_2q_1 & \frac{3}{4}p_1p_2 + \frac{1}{4}q_1q_2 & \frac{3}{4}p_1q_1 + \frac{1}{4}p_2q_2 & \hat{\nu} - \frac{3}{4}p_1^2 + \frac{1}{4}q_2^2 \\ -\frac{1}{4}p_1p_2 - \frac{3}{4}q_1q_2 & -\frac{3}{4}p_1q_2 + \frac{5}{4}p_2q_1 & -\hat{\nu} - \frac{1}{4}p_2^2 + \frac{3}{4}q_1^2 & -\frac{3}{4}p_1q_1 - \frac{1}{4}p_2q_2 \end{vmatrix}, \tag{4.3}$$

$$(p_i, q_i) \equiv (p_i^0, q_i^0), \quad \hat{\nu} = \nu + \frac{1}{8}E,$$

$$D_2 = M_{1212} + 2M_{1234} + M_{3434} \tag{4.4a}$$

$$= 2\nu^2 - (1/32)E^2 + (15/8)M^2, \tag{4.4b}$$

and $M_{pqrs}$ is that minor of the determinant $D_0$ containing those elements common to rows $p, q$ and columns $r, s$ (note that $M_{3412} = M_{1234}$).

### 4.1. *Planar motion*

Substituting the planar solution (3.4) and (3.5) into (4.3), we find that (4.2) can be factored to obtain

$$(\lambda + \alpha)^2 = -M_{1212} = \frac{1}{4}E^{-1}(1 - \alpha^2 E)^{1/2} \times \left[\pm E^{3/2} - 4(1 - \alpha^2 E)^{1/2}\right] \tag{4.5a}$$

and

$$(\lambda + \alpha)^2 = -M_{3434} = -\frac{3}{4}E^{-1}(1 - \alpha^2 E)^{1/2} \times \left[\pm E^{3/2} + \frac{4}{3}(1 - \alpha^2 E)^{1/2}\right], \tag{4.5b}$$

wherein the alternative signs are vertically ordered with those in (3.4) and (3.5a). The first of these,

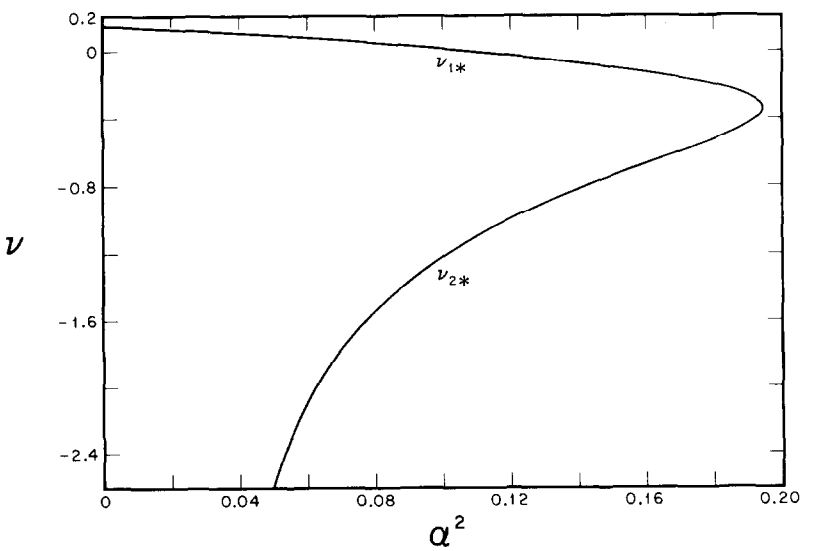



Fig. 7. The Hopf-bifurcation points $\nu_{1*}$ and $\nu_{2*}$, as determined by the vanishing of the discriminant (4.6c).

(4.5a), corresponds to a planar perturbation for which $p_2$ and $q_2$ retain their null values. The second, (4.5b), corresponds to a non-planar perturbation for which $p_1$ and $q_1$ retain the values (3.5a, b).

Both (4.5a) and (4.5b) yield $\mathrm{Re}\lambda \le 0$ on each of branches I ($-\infty < \nu \le \nu_3$), $\mathrm{III}_1$ ($\nu_1 \le \nu \le \nu_2$) and $\mathrm{III}_3$ ($\nu \ge \nu_4$), which therefore are stable (more precisely, loci of stable singular points); (4.5a) and (4.5b) yield $\mathrm{Re}\lambda > 0$ on branches II ($\nu_1 < \nu < \nu_3$) and $\mathrm{III}_2$ ($\nu_2 < \nu < \nu_4$), respectively, which therefore are unstable. But note that if the motion is constrained to be planar – i.e. if $p_2 = q_2 \equiv 0$ – all of branch III is stable. We recall (see section 3) that the bifurcation points $\nu_1$ and $\nu_3$ disappear for $\alpha > 0.433$ and that the bifurcation points $\nu_2$ and $\nu_4$ disappear for $\alpha > 0.625$. It follows that branches II and $\mathrm{III}_2$ disappear for $\alpha > 0.433$ and $\alpha > 0.625$, respectively, and that the entire planar resonance curve is single valued and stable for $\alpha > 0.625$.

### 4.2. *Non-planar motion*

The necessary and sufficient conditions for the real parts of the roots of the quartic equation (4.2) to have non-positive real parts – i.e. for the perturbation (4.1) be stable – are [10]

$$F(0) = \alpha^4 + D_2\alpha^2 + D_0 \ge 0, \tag{4.6a}$$

$$D_2 + 2\alpha^2 \ge 0, \quad \left(D_2 + 4\alpha^2\right)^2 - 4D_0 \ge 0. \tag{4.6b,c}$$

The discriminant $F(0)$ is negative on branch V ($\nu_4 < \nu < \nu_5$), as well as on branches II and $\mathrm{III}_2$ (see above). The discriminant (4.6c) is negative on branch IV for $\nu_{1*} < \nu < \nu_{2*}$ ($\mathrm{IV}_2$, unstable); all of (4.6a, b, c) are satisfied for either $\nu_2 < \nu < \nu_{1*}$ ($\mathrm{IV}_1$, stable) or for $\nu_{2*} < \nu < \nu_5$ ($\mathrm{IV}_3$, stable). The numerical solution of $(D_2 + 4\alpha^2)^2 = 4D_0$ is plotted in fig. 7. The bifurcation points $\nu_{1*}$ and $\nu_{2*}$ coalesce for $\alpha = 0.441$, and branch $\mathrm{IV}_2$ disappears for $\alpha > 0.441$; for $\alpha \ll 1$,

$$\nu_{1*} = -\frac{1}{8}\alpha^{-2} + 3.638\alpha^4 + \mathcal{O}(\alpha^{10}), \tag{4.7a}$$

$$\nu_{2*} = 0.154 - 1.153\alpha^2 + \mathcal{O}(\alpha^4). \tag{4.7b}$$

Table I
The numbers of stable and unstable equilibrium points (see sections 3 and 4)

| $\alpha$ interval | $\nu$ interval | Stable | Unstable |
|---|---|---|---|
| 0, 0.374 | $-\infty, \nu_1$ | 1 | 0 |
| | $\nu_1, \nu_2$ | 2 | 1 |
| | $\nu_2, \nu_{1*}$ | 3 | 2 |
| | $\nu_{1*}, \nu_3$ | 1 | 4 |
| | $\nu_3, \nu_{2*}$ | 0 | 3 |
| | $\nu_{2*}, \nu_4$ | 2 | 1 |
| | $\nu_4, \nu_5$ | 3 | 2 |
| | $\nu_5, \infty$ | 1 | 0 |
| 0.374, 0.433 | $-\infty, \nu_1$ | 1 | 0 |
| | $\nu_1, \nu_2$ | 2 | 1 |
| | $\nu_2, \nu_3$ | 3 | 2 |
| | $\nu_3, \nu_{1*}$ | 2 | 1 |
| | $\nu_{1*}, \nu_{2*}$ | 0 | 3 |
| | $\nu_{2*}, \nu_4$ | 2 | 1 |
| | $\nu_4, \nu_5$ | 3 | 2 |
| | $\nu_5, \infty$ | 1 | 0 |
| 0.433, 0.441 | $-\infty, \nu_2$ | 1 | 0 |
| | $\nu_2, \nu_{1*}$ | 2 | 1 |
| | $\nu_{1*}, \nu_{2*}$ | 0 | 3 |
| | $\nu_{2*}, \nu_4$ | 2 | 1 |
| | $\nu_4, \nu_5$ | 3 | 2 |
| | $\nu_5, \infty$ | 1 | 0 |
| 0.441, 0.526 | $-\infty, \nu_2$ | 1 | 0 |
| | $\nu_2, \nu_4$ | 2 | 1 |
| | $\nu_4, \nu_5$ | 3 | 2 |
| | $\nu_5, \infty$ | 1 | 0 |
| 0.526, 0.625 | $-\infty, \nu_2$ | 1 | 0 |
| | $\nu_2, \nu_4$ | 2 | 1 |
| | $\nu_4, \infty$ | 1 | 0 |
| 0.625, $\infty$ | $-\infty, \infty$ | 1 | 0 |

The bifurcation point $\nu_5$ is given by (3.14); we recall that $\nu_5$ disappears for $\alpha > 0.526$. Note that there is at least one stable equilibrium point for all $\nu$ if $\alpha > 0.441$ and that there is only one such point if $\alpha > 0.526$.

We remark that $\nu_{1,2,3,4,5}$ are all bifurcation points in Poincaré's sense [11], which requires that: (i) there be a transition from stability to instability; (ii) $F(0) = 0$ (i.e. $\lambda = 0$); (iii) there be a change in the number of equilibrium points. Conditions (ii) and (iii) are not satisfied at $\nu = \nu_{1*,2*}$, which

are Hopf-bifurcation points, at which the real parts of a pair of complex-conjugate roots become positive.

The preceding results are summarized in table I and illustrated in figs. 1–4.

## 5. Asymptotic trajectories

The asymptotic (i.e. attained after the decay of transients) phase-space trajectory depends on $\alpha$ and $\nu$ and, in a more limited sense, the initial conditions. Numerical integrations suggest that, with possible exceptions (see penultimate paragraph in section 6), a trajectory terminates on a stable equilibrium point if such exists for the prescribed $\alpha$ and $\nu$ (see figs. 1–4 and table I); if there are more than one (either two or three in the present context) such points, which is attained depends on the initial conditions. The asymptotic trajectories in those parametric domains in which no stable equilibrium points exist are non-planar and either periodic or chaotic†. The periodic trajectories are limit cycles and are uniquely determined by $\alpha$ and $\nu$ up to the reflection $(p_2, q_2) \rightarrow -(p_2, q_2)$, and which of a complementary pair is realized depends on the initial conditions; e.g. the limit cycle obtained for the initial conditions $(p_1, q_1, p_2, q_2) = (0, 0, 0, 1)$ is replaced by its complement if the initial conditions are replaced by either $(0, 0, 0, -1)$ or $(0, 0, 1, 0)$.

Periodic solutions of (2.13) presumably could be obtained by positing Fourier series and regarding the fundamental frequency as an eigenvalue; however, the algebra is forbidding. The angular frequency (for the slow time $\tau$) in the limit $\nu \uparrow \nu_{2*}$ (we anticipate that the Hopf bifurcation at $\nu = \nu_{2*}$ is supercritical, so that the periodic solution is stable), as obtained by setting $\lambda = i\omega_*$ in (4.2) and requiring the imaginary part of the result to vanish (the vanishing of the real part is equivalent to the vanishing of the left-hand side of (4.6c)), is given by

$$\omega_*^2 = \alpha^2 + \frac{1}{2}D_2 = (11/96)E^2 + (5/6)\beta E - (2/3)(\alpha^2 + \beta^2). \tag{5.1}$$

[The second pair of eigenvalues for $\nu = \nu_{2*}$ is given by $\lambda = -2\alpha \pm i\omega_*$.] Starting from (4.7b) and perturbing (3.3b) as $\alpha^2 \downarrow 0$, we obtain

$$\omega_* = 1.183 - 2.845\alpha^2 + \mathcal{O}(\alpha^4). \tag{5.2}$$

A similar calculation could be carried out for $\nu \downarrow \nu_{1*}$ but would be significant only for $0.374 < \alpha < 0.441$ (in which range $\nu_{1*} > \nu_3$).

Chaotic solutions are not uniquely determined by finite numerical specifications of $\alpha$, $\nu$ and the initial conditions. Nevertheless, their general appearance (or at least that of their $p_1$–$p_2$ projections) does appear to be determined for specified $\alpha$ and $\nu$ to within a possible mixing of a complementary pair of trajectories that are related through the reflection $(p_2, q_2) \rightarrow -(p_2, q_2)$; see below.

## 6. Numerical integrations

Numerical integrations of (2.13) were carried out using an Adams–Bashforth routine with a round-off error of less than $10^{-4}$. (Decreasing the round-off error from $10^{-4}$ to $10^{-6}$ made no perceptible difference in a sub-set of the results presented here.) Various plane projections $(p_1$–$p_2, q_1$–$q_2, p_1$–$q_1, p_2$–$q_2)$ and Poincaré sections of the phase-space trajectories were examined, after which it was decided that the $p_1$–$p_2$ projections offered the optimum combination of information and computing economy. We recall (section 2) that these projections are maps of the $x$–$y$ projection of the pendulum-bob trajectories at the periodically spaced instants $\omega t = 0 \pmod{2\pi}$.

The power spectra of $N$-point runs of $E$ were determined through a fast-Fourier-transform pro-

†It follows from the Poincaré–Bendixson theorem that a planar trajectory must be asymptotic to either a fixed point or a periodic limit cycle. Neither planar limit cycles nor non-planar, quasi-periodic (comprising two or more incommensurate frequencies) asymptotic trajectories were obtained over the parametric domains that were numerically explored, but I have not demonstrated the non-existence of the latter.

gram according to

$$P(f_k) = (2\Delta/N)\Big|\sum_{n=0}^{N-1} E(\tau_n)w(\tau_n) \times \exp(-2\pi i f_k \tau_n)\Big|^2, \qquad (6.1)$$

where

$$f_k = k/T \quad \text{and} \quad \tau_n = n\Delta \qquad (6.2a,b)$$

are the discrete, dimensionless frequency and time, $\Delta$ is the increment of $\tau$, $T \equiv N\Delta$ is the length of

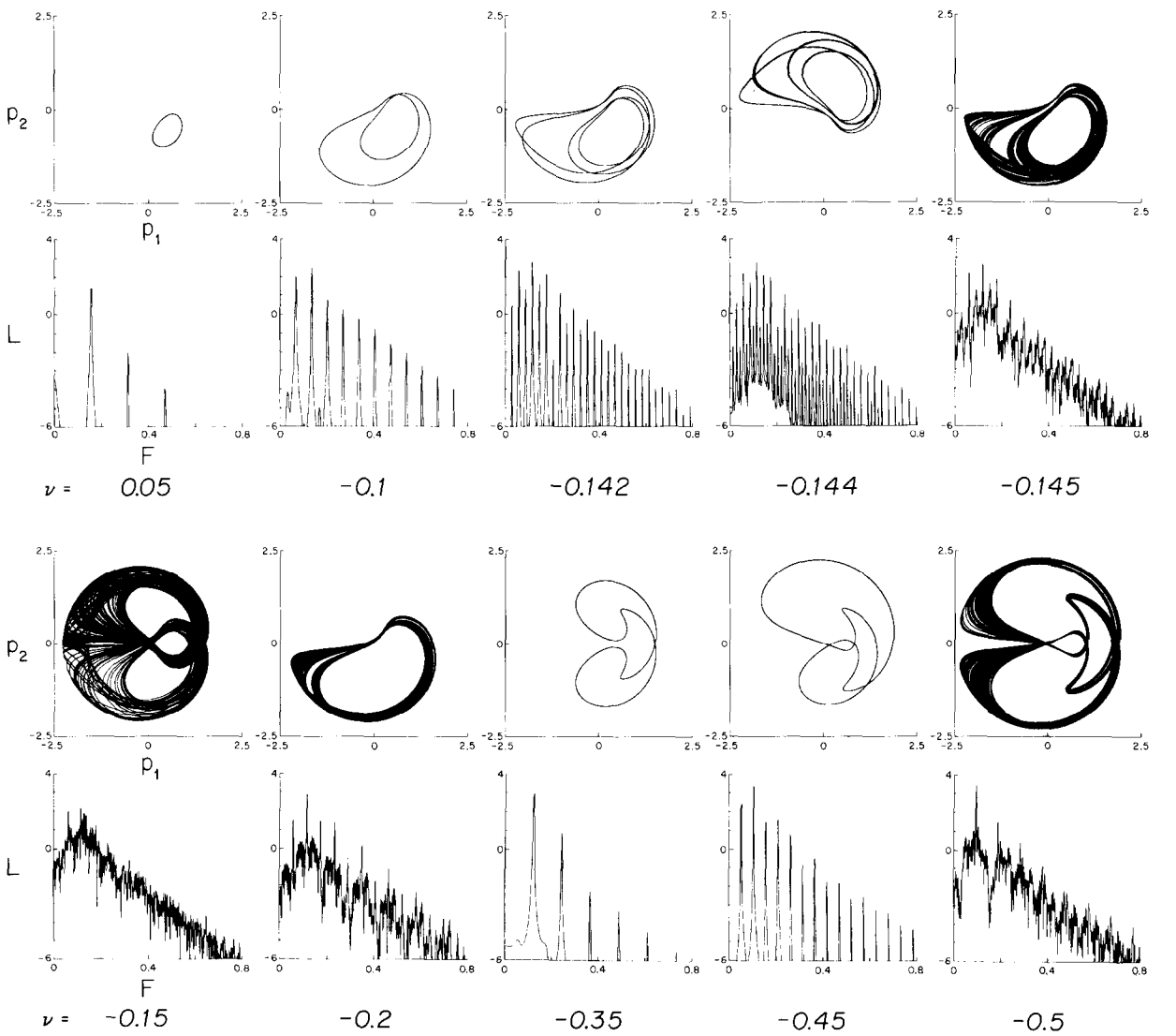



Fig. 8. Phase-plane ($p_1$–$p_2$) trajectories and power spectra ($L \equiv \log_{10} P$ vs. $f$) for $\alpha = 1/4$ and $\nu = 0.05$, $-0.10$, $-0.142$, $-0.144$, $-0.145$, $-0.15$, $-0.20$, $-0.35$, $-0.45$ and $-0.50$. The initial conditions are $(p_1, q_1, p_2, q_2) = (0,0,0,1)$ except for $\nu = -0.144$, where they were taken as $(0,0,1,0)$ in order to illustrate the reflection $(p_2, q_2) \rightarrow -(p_2, q_2)$ (see text). The zero-frequency component of the power spectrum has been eliminated.

the run ($0 < \tau < T$), and

$$w(\tau) = (2/3)^{1/2}[1 - \cos(2\pi\tau/T)]$$
$$= (8/3)^{1/2}\sin^2(\pi\tau/T) \qquad (6.3)$$

is the window (or taper) function, normalized to an r.m.s. value of unity. The value of $\Delta$ for all runs was $2^{-4}$. The corresponding Nyquist frequency, 16, is roughly two orders of magnitude above the fundamental frequency $f_1$ or spectral peak in the following results. Numerical noise was below $P = 10^{-8}$. The zero-frequency components of the power spectra in fig. 8 have been eliminated by omitting the first two ($k = 0, 1$) terms.

The parametric domain of principal interest for the numerical integrations is either $0 < \alpha < 0.374$ and $\nu_3 < \nu < \nu_{2*}$ or $0.374 < \alpha < 0.441$ and $\nu_{1*} < \nu < \nu_{2*}$, in which there are no stable equilibrium points and the phase-space trajectories are either periodic or chaotic (see section 5). Runs were made through these $\nu$-intervals with increments of $\Delta\nu \leq 0.05$ for $\alpha = 0.1$, 0.25 and 0.4. All trajectories for $\alpha = 0.4$ and $\nu_{1*} = -0.667 < \nu < -0.128 = \nu_{2*}$ were asymptotically periodic and qualitatively similar to the periodic trajectories for $\alpha = 0.25$ and either $\nu = 0.05$ or $-0.35$ (see fig. 8). The trajectories for $\alpha = 0.1$ were qualitatively similar to those for $\alpha = 0.25$, although the transients were longer (as expected), and the relative density of periodic trajectories in $\nu_3 < \nu < \nu_{2*}$ was smaller. A few runs were made for $\alpha = 0$ and $\nu_3 = -0.945 < \nu < 0.154 = \nu_{2*}$ and yielded only what appeared to be chaotic trajectories.

The results for $\alpha = 1/4$ are summarized in table II and fig. 8, with $\nu$ decreasing from above $\nu_{2*} = 0.072$ to below $\nu_3 = -0.892$. The data for $0 < \tau < 2^8$ are omitted from the $p_1$–$p_2$ phase-plane plots in order to eliminate transients; these transients typically decayed in $\tau < 2^7$, but that for $\nu = -0.06$ persisted into the interval $(2^9, 2^{10})$. The initial conditions are $(p_1, q_1, p_2, q_2) = (0, 0, 0, 1)$ except as noted.

Trajectories for $0.09 \geq \nu > 0.072$ spiral slowly in to the calculated equilibrium points. The fundamental frequencies of the damped oscillations for $\nu = 0.09$, 0.08 and 0.075 are $f_1 = 0.162$, 0.160 and 0.160, respectively, which compare with $f_* = 0.160$ given by (5.2) at $\nu = \nu_{2*} = 0.072$.

The trajectories for $0.072 > \nu > -0.04$, e.g. $\nu = 0.05$ in fig. 8, are simple limit cycles that project as ovals in the $p_1$–$p_2$ plane and comprise $f_1$ and its harmonics, with $f_1$ decreasing from $f_*$ with decreasing $\nu$; see fig. 9. The stability of these limit cycles for $\nu$ just below the Hopf-bifurcation point $\nu_{2*}$, together with the absence of limit cycles just above $\nu_{2*}$, strongly suggests that the bifurcation is supercritical.

Very weak period doubling is observed at $\nu = -0.04$, and the half-order subharmonics increase in strength as $\nu$ is decreased from $-0.04$ to $-0.144$; see $\nu = -0.10$ in fig. 8. Period quadrupling (i.e. a second period doubling) first occurs as $\nu$ is decreased through the interval $(-0.13, -0.14)$, and the quarter-order subharmonics increase in strength down to $\nu = -0.144$; see $\nu = -0.142$ in fig. 8. Period octupling (a third period doubling) is observed at $\nu = -0.144$; see fig. 8. The metamorphosis of the trajectories from almost elliptical ovals through increasing convolutions at each period doubling is illustrated by the results for $\nu = 0.05$, $-0.10$, $-0.142$, $-0.144$ and $-0.145$ in fig. 8 after allowing for the fact that the initial conditions for $\nu = -0.144$ were altered to obtain the complement of the trajectory that had been obtained for initial conditions identical with those for $\nu = 0.05$, $-0.10$ and $-0.142$.

Chaotic trajectories occur throughout the $\nu$ interval $[-0.145, -0.20]$. The typical $p_1$–$p_2$ trajectory within this interval, e.g. $\nu = -0.15$ (see also $\nu = -0.5$) in fig. 8, appears to be symmetric with respect to $p_2 = 0$ (see third paragraph in section 5) after a sufficiently long time, although it may spend substantial intervals in either the top or bottom half of this pattern, transferring from one to the other half at aperiodically spaced times that are quite sensitive to the initial conditions. The available evidence (broadband spectra, sensitivity to initial conditions, and intermittent transfer between two complementary domains of the phase space) strongly suggest that these trajectories lie

Table II

Numerical results for $\alpha = 1/4$. The initial conditions were $(p_1, q_1, p_2, q_2) = (0,0,0,1)$ except as noted. The total number of points in a run was $2^4T$. The fundamental frequency $f_1$ is identified by the spectral line of maximum amplitude

| $\nu$ | $T$ | $p_1$ vs. $p_2$ | Spectrum (above $P = 10^{-5}$) |
|---|---|---|---|
| 0.09, 0.08, 0.075 | $2^9$ | FP* | $f_1$ (transient) |
| 0.05 | $2^9$ | LC* | $f_1(1,2,3)$ |
| 0, −0.02, −0.04 | $2^9$ | LC | $f_1(1,2,3,4)$ |
| −0.05† | $2^{10}$ | LC | $f_1(1,2,\ldots,5)$; $\frac{1}{2}f_1$ |
| −0.06† | $2^{11}$ | LC | $f_1(1,2,\ldots,5)$; $\frac{1}{2}f_1(1,3,5,7)$ |
| −0.07†, −0.08 | $2^9$ | LC | $f_1(1,2,\ldots,5)$; $\frac{1}{2}f_1(1,3,\ldots,9)$ |
| −0.10, −0.11, −0.12, −0.13 | $2^{10}$ | LC | $f_1(1,2,\ldots,6)$; $\frac{1}{2}f_1(1,3,\ldots,13)$ |
| −0.14 | $2^{10}$ | LC | $f_1(1,2,\ldots,5)$; $\frac{1}{2}f_1(1,3,\ldots,11)$; $\frac{1}{4}f_1(1,3,5,9,11,13)$ |
| −0.142 | $2^{10}$ | LC | $f_1(1,2,\ldots,6)$; $\frac{1}{2}f_1(1,3,\ldots,11)$; $\frac{1}{4}f_1(1,3,\ldots,25)$ |
| −0.143† | $2^{10}$ | LC | $f_1(1,2,\ldots,6)$; $\frac{1}{2}f_1(1,3,\ldots,13)$; $\frac{1}{4}f_1(1,3,\ldots,27)$ |
| −0.144†† | $2^{10}$ | LC | $f_1(1,2,\ldots,6)$; $\frac{1}{2}f_1(1,3,\ldots,13)$; $\frac{1}{4}f_1(1,3,\ldots,25)$; $\frac{1}{8}f_1(1,3,\ldots,35)$ |
| −0.145 | $2^{10}$ | SA* | $f_1(1,2,3)$; $\frac{1}{2}f_1(1,3,5)$; $\frac{1}{4}f_1(1,3,5)$; BB* |
| −0.15 | $2^{11}$ | SA | BB |
| −0.16 | $2^9$ | SA | $f_1(1,2,3,4)$; $\frac{1}{2}f_1(1,3,5,7)$; BB |
| −0.17 | $2^9$ | SA | $f_1$; $\frac{1}{2}f_1(1,3)$; BB |
| −0.18 | $2^9$ | SA | $f_1$; $\frac{1}{2}f_1$; BB |
| −0.19 | $2^9$ | SA | BB |
| −0.20 | $2^{10}$ | SA | $f_1(1,2,\ldots,5)$; $\frac{1}{2}f_1(1,3,\ldots,11)$; BB |
| −0.21 | $2^9$ | LC | $f_1(1,2,\ldots,7)$; $\frac{1}{2}f_1(1,3,\ldots,13)$ |
| −0.22 | $2^9$ | LC | $f_1(1,2,\ldots,7)$ |
| −0.23, −0.24 | $2^9$ | SA | $f_1(1,2)$; BB |
| −0.25 | $2^{10}$ | LC | $f_1(1,2,\ldots,7)$; $\frac{1}{2}f_1(1,3,\ldots,15)$ |
| −0.26† | $2^{10}$ | LC | $f_1(1,2,\ldots,7)$ |
| −0.27† | $2^{10}$ | LC | $f_1(1,2,\ldots,6)$ |
| −0.28†, −0.29, −0.30, −0.35 | $2^{10}$ | LC | $f_1(1,2,\ldots,5)$ |
| −0.40 | $2^{10}$ | LC | $f_1(1,2,\ldots,6)$; $\frac{1}{2}f_1$ |
| −0.45 | $2^{10}$ | LC | $f_1(1,2,\ldots,8)$; $\frac{1}{2}f_1(1,3,\ldots,15)$ |
| −0.50 | $2^{11}$ | SA | $f_1(1,2,\ldots,5)$; $\frac{1}{2}f_1(1,3,\ldots,9)$; BB |
| −0.52 | $2^9$ | SA | $f_1$; BB |
| −0.54, −0.56, −0.58 | $2^9$ | LC | $f_1(1,2,\ldots,9)$ |
| −0.60 | $2^9$ | LC | $f_1(1,2,\ldots,9)$ |
| −0.63, −0.65 | $2^9$ | LC | $f_1(1,2,\ldots,11)$ |
| −0.70 | $2^9$ | LC | $f_1(1,2,\ldots,12)$ |
| −0.75 | $2^9$ | LC | $f_1(1,2,\ldots,13)$; $\frac{1}{2}f_1(1,3,\ldots,15)$ |
| −0.78 | $2^9$ | SA | $f_1(1,2)$; BB |
| −0.80 | $2^{10}$ | SA | $f_1(1,2,\ldots,15)$; BB |
| −0.85, −0.90 | $2^{10}$ | SA | $f_1(1,2)$; BB |
| −0.92 | $2^{10}$ | LC | $f_1(1,2,\ldots,17)$; $\frac{1}{2}f_1(1,3,\ldots,35)$; $\frac{1}{4}f_1(1,3,\ldots,59)$ |
| −0.95 | $2^9$ | FP | — |

†Initial conditions: $(p_1, q_1, p_2, q_2) = (0, 1, 1, 0)$. ††Initial conditions: $(p_1, q_1, p_2, q_2) = (0,0,1,0)$. *BB ≡ broadband; FP ≡ fixed (equilibrium) point; LC ≡ periodic limit cycle; SA ≡ chaotic trajectory.

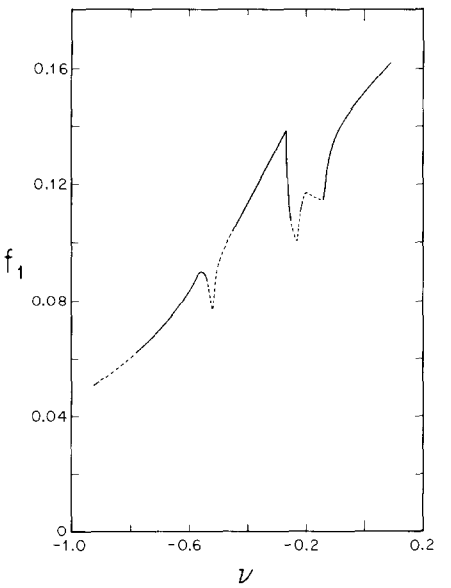



Fig. 9. The fundamental frequency $f_1$ of the periodic trajectories for $\alpha = 1/4$. The dashed intervals are those in which chaotic trajectories were obtained.

on a fractal attractor of Lorenz's type. The trajectories for $\nu = -0.145$ and $-0.20$ (see fig. 8) are exceptional in being confined to one of the two putative domains of the attractor, which then appears to be of Rössler's type. The complement of the trajectory shown in fig. 8 for $\nu = -0.145$ was obtained by altering the initial conditions from $(p_1, q_1, p_2, q_2) = (0,0,0,1)$ to $(0,0,1,0)$; on the other hand, essentially the same trajectory as that shown for $\nu = -0.20$ was obtained with each of the three sets of initial conditions $(0,0,0,1)$, $(0,0,1,0)$ and $(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

The trajectories for $\nu = -0.21$ and $-0.22$ are periodic, with period doubling as $\nu$ increases from $-0.22$ to $-0.21$. Those for $\nu = -0.23$ and $-0.24$ are chaotic and appear to lie on fractal attractors of Lorenz's type. Those for $-\nu = 0.25$ (.01) 0.30 (.05) 0.45 are periodic, with period doubling at $-0.25$ as $\nu$ increases to $-0.24$ and at $-0.40$ and $-0.45$ as $\nu$ decreases to $-0.50$, where a fractal attractor (presumably) of Lorenz's type again occurs; see $\nu = -0.35$, $-0.45$ and $-0.50$ in fig. 8. It is probable that period quadrupling and octupling would be observed through finer sampling over each of the spectral intervals that separate those of chaotic trajectories, just as in the $\nu$ interval $(-0.13, -0.145)$. Moreover, the appearance of fractal attractors within the narrow interval $(-0.22, -0.25)$ suggests that there could be additional such subintervals within the interval $(-0.30, -0.50)$. The $\nu$ interval $(-0.50, -0.95)$ also contains sub-intervals of both periodic and chaotic trajectories.

The apparently chaotic trajectory obtained for $\nu = -0.90$ could be transient (or metastable [13]) and could terminate at the planar equilibrium point just below $\nu_3 = -0.892$ after a sufficiently long time. The limit cycle obtained for $\nu = -0.92$ also could be transient, but it seems more likely in this case that the limit cycle also has a basin of attraction that contains $(0,0,0,1)$. The stable equilibrium points on branch I were attained for both $\nu = -0.92$ and $-0.90$ with the initial conditions $(1,0,0,10^{-2})$.

The frequency $f_1$, as inferred from the peak spectral lines in the power spectra for the periodic trajectories, is plotted in fig. 9. The curve appears to be smooth except in the $\nu$ intervals $(-0.145, -0.27)$ and $(-0.45, -0.56)$, within which chaotic trajectories were obtained (but it remained smooth in $[-0.78, -0.90]$ despite the chaotic trajectories in that interval).

## Appendix A

*Pendulum-bob trajectory*

Starting from (2.6), we let

$$\hat{x} = x\cos\phi + y\sin\phi \equiv \varepsilon^{1/3} lA\cos(\theta - \psi) \qquad \text{(A.1a)}$$

and

$$\hat{y} = -x\sin\phi + y\cos\phi \equiv \varepsilon^{1/3} lB\sin(\theta - \psi), \qquad \text{(A.1b)}$$

where $\phi$ is the slowly varying angle between the major $(\hat{x})$ axis of the ellipse and the original

$x$-axis, $\varepsilon^{1/3}l(A, B)$ are the slowly varying semi-(major, minor) axes, and $\psi$ is a slowly varying angle. Substituting (2.6) into (A.1) and equating coefficients of $\cos\theta$ and $\sin\theta$, respectively, we obtain

$$p_1 = A\cos\phi\cos\psi + B\sin\phi\sin\psi, \tag{A.2a}$$

$$q_1 = A\cos\phi\sin\psi - B\sin\phi\cos\psi, \tag{A.2b}$$

$$p_2 = A\sin\phi\cos\psi - B\cos\phi\sin\psi, \tag{A.2c}$$

$$q_2 = A\sin\phi\sin\psi + B\cos\phi\cos\psi, \tag{A.2d}$$

$$\frac{A}{B} = \left[\frac{1}{2}E \pm \frac{1}{2}(E^2 - 4M^2)^{1/2}\right]^{1/2}, \tag{A.3}$$

$$E = A^2 + B^2 \equiv R^2, \quad M = AB, \tag{A.4a,b}$$

$$\tan 2\phi = 2(p_1p_2 + q_1q_2)/(p_1^2 + q_1^2 - p_2^2 - q_2^2), \tag{A.5}$$

and

$$\tan\psi = (Aq_2 - Bp_1)/(Ap_2 + Bq_1), \tag{A.6}$$

where $E$ and $M$ are the dimensionless energy and momentum (2.10).

## Appendix B

*The bounding hypersphere*

Multiplying (2.13a, b, c, d) through by $(p_1, q_1, p_2, q_2)$, summing, and invoking (2.10a) for $R^2$, we obtain

$$\frac{R\,\mathrm{d}R}{\mathrm{d}\tau} = q_1 - \alpha R^2, \tag{B.1}$$

which admits the implicit solution

$$R = R_0 e^{-\alpha\tau} + \int_0^\tau \frac{q_1(\eta)}{R(\eta)} e^{-\alpha(\tau-\eta)}\mathrm{d}\eta. \tag{B.2}$$

Invoking $|q_1/R| \equiv |q_1|/(p_1^2 + q_1^2 + p_2^2 + q_2^2)^{1/2} \le 1$, we infer that

$$|R - R_0 e^{-\alpha\tau}| \le \int_0^\tau e^{-\alpha(\tau-\eta)}\mathrm{d}\eta = \frac{1 - e^{-\alpha\tau}}{\alpha}, \tag{B.3}$$

which, in turn, implies the asymptotic bound

$$R \le 1/\alpha \quad (\alpha\tau \uparrow \infty). \tag{B.4}$$

We next rewrite (B.1) in the form

$$\frac{R\,\mathrm{d}R}{\mathrm{d}\tau} = \alpha(R_*^2 - \hat{R}^2), \tag{B.5}$$

where

$$\hat{R}^2 = p_1^2 + (q_1 - R_*)^2 + p_2^2 + q_2^2, \tag{B.6a}$$

$$R_* = (2\alpha)^{-1}. \tag{B.6b}$$

It follows from (B.5) that $\mathrm{d}R/\mathrm{d}\tau \gtrless 0$ in the interior/exterior of $S_*$, a hypersphere of radius $R_*$ with center at $(0, R_*, 0, 0)$ that is contained by the hypersphere of the preceding paragraph, $R = 2R_*$ (cf. Lorenz [3]), but it does not follow (as was incorrectly stated in [1]) that the asymptotic trajectories for non-planar motion must lie within $S_*$.

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